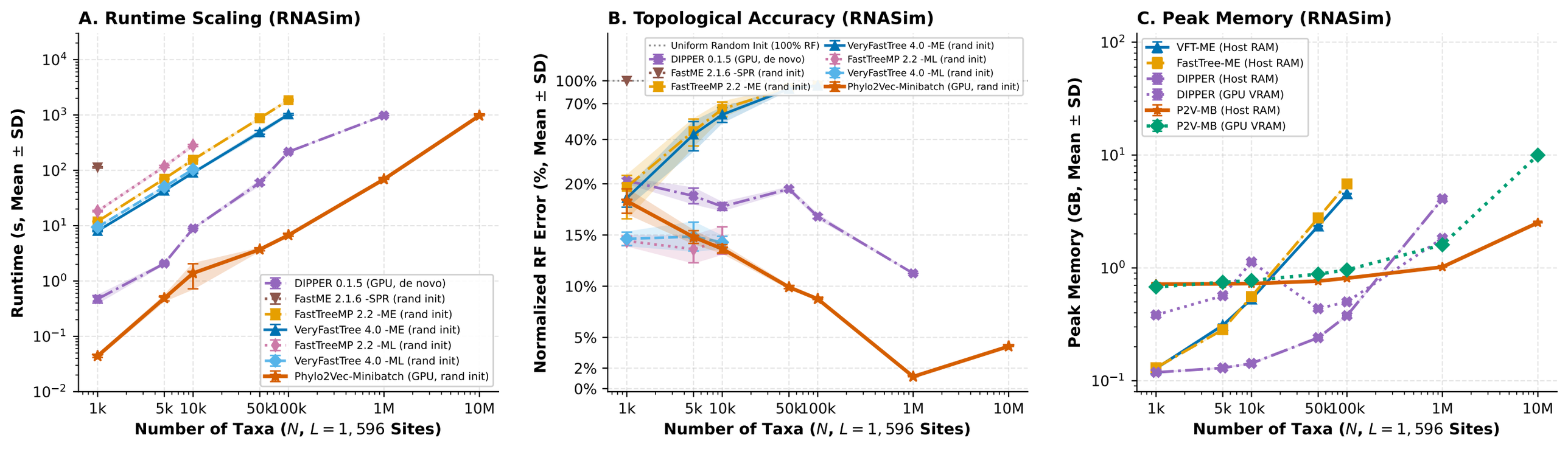
A. Runtime Scaling (RNASim)
Runtime (s, Mean ± SD)
Number of Taxa (N, L = 1,596 Sites)
DIPPER 0.1.5 (GPU, de novo)
FastME 2.1.6 -SPR (rand init)
FastTreeMP 2.2 -ME (rand init)
VeryFastTree 4.0 -ME (rand init)
FastTreeMP 2.2 -ML (rand init)
VeryFastTree 4.0 -ML (rand init)
Phylo2Vec-Minibatch (GPU, rand init)
B. Topological Accuracy (RNASim)
Normalized RF Error (%, Mean ± SD)
Number of Taxa (N, L = 1,596 Sites)
Uniform Random Init (100% RF)
DIPPER 0.1.5 (GPU, de novo)
FastME 2.1.6 -SPR (rand init)
FastTreeMP 2.2 -ME (rand init)
VeryFastTree 4.0 -ME (rand init)
FastTreeMP 2.2 -ML (rand init)
VeryFastTree 4.0 -ML (rand init)
Phylo2Vec-Minibatch (GPU, rand init)
C. Peak Memory (RNASim)
Peak Memory (GB, Mean ± SD)
Number of Taxa (N, L = 1,596 Sites)
VFT-ME (Host RAM)
FastTree-ME (Host RAM)
DIPPER (Host RAM)
DIPPER (GPU VRAM)
P2V-MB (Host RAM)
P2V-MB (GPU VRAM)